



## PES UNIVERSITY

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Department of Computer Science & Engineering

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**UE24MA242A: Mathematics for Computer Science Engineers**

### Classwork Problems

| #  | UNIT_2 Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Exercise Number                                     |
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| 1. | <p>Let <math>X_1, X_2, X_3</math> be independent, each with unknown mean <math>\mu</math> and known variance <math>\sigma^2 = 1</math>.</p> <p>Consider the estimator:</p> $\hat{\mu}_4 = \frac{2X_1 - X_2 + 3X_3}{6}.$ <p>Find:</p> <ol style="list-style-type: none"> <li>1. The bias of <math>\hat{\mu}_4</math></li> <li>2. The variance of <math>\hat{\mu}_4</math></li> <li>3. The mean squared error (MSE) of <math>\hat{\mu}_4</math></li> </ol> <p>Let <math>\hat{\mu}_4 = \frac{2X_1 - X_2 + 3X_3}{6}</math> with independent <math>X_i</math> where <math>E[X_i] = \mu</math>, <math>\text{Var}(X_i) = \sigma^2 = 1</math>.</p> <ul style="list-style-type: none"> <li>• Bias <math display="block">E[\hat{\mu}_4] = \frac{2\mu - \mu + 3\mu}{6} = \frac{4\mu}{6} = \frac{2}{3}\mu \Rightarrow \text{Bias}(\hat{\mu}_4) = E[\hat{\mu}_4] - \mu = -\frac{\mu}{3}.</math> </li> <li>• Variance (independence): <math>\rightarrow \text{Var}(\hat{\mu}_4) = \left(\frac{2}{6}\right)^2 \sigma^2(X_1) + \left(-\frac{1}{6}\right)^2 \sigma^2(X_2) + \left(\frac{3}{6}\right)^2 \sigma^2(X_3)</math> <math display="block">\text{Var}(\hat{\mu}_4) = \left(\frac{2}{6}\right)^2 + \left(\frac{-1}{6}\right)^2 + \left(\frac{3}{6}\right)^2 = \frac{1}{9} + \frac{1}{36} + \frac{1}{4} = \frac{7}{18}.</math> </li> <li>• MSE <math display="block">\text{MSE}(\hat{\mu}_4) = \text{Var}(\hat{\mu}_4) + \{\text{Bias}(\hat{\mu}_4)\}^2 = \frac{7}{18} + \left(\frac{\mu}{3}\right)^2 = \frac{7}{18} + \frac{\mu^2}{9}.</math> </li> </ul> <p>So, Bias = <math>-\mu/3</math>, Variance = <math>7/18</math>, and MSE = <math>7/18 + \mu^2/9</math>.</p> | Principles of Point Estimation, Introduction to MSE |

2.

Let  $X \sim \text{Binomial}(n = 10, p)$ . Consider the estimator

$$\hat{p}_2 = \frac{X + 1}{12}.$$

Mean Squared Error for Binomial,

(a) Find the bias of  $\hat{p}_2$ .

(b) Find the variance of  $\hat{p}_2$ .

© Find the mean squared error (MSE) of  $\hat{p}_2$ .

**(a) Bias of  $\hat{p}_2$**

First compute the expectation:

$$E[\hat{p}_2] = \frac{\overset{\rightarrow n p}{E[X]} + 1}{12} = \frac{10p + 1}{12}.$$

Bias:

$$\text{Bias}(\hat{p}_2) = E[\hat{p}_2] - p = \frac{10p + 1}{12} - p = \frac{10p + 1 - 12p}{12} = \frac{1 - 2p}{12}.$$

**(b) Variance of  $\hat{p}_2$**

Since expectation shift (+1/12) doesn't affect variance:

$$\text{Var}(\hat{p}_2) = \frac{1}{12^2} \text{Var}(X).$$

For Binomial(10, p):

$$\overset{\rightarrow n p q}{\text{Var}(X)} = 10p(1 - p).$$

So,

$$\text{Var}(\hat{p}_2) = \frac{10p(1 - p)}{144} = \frac{5p(1 - p)}{72}.$$

**(c) Mean Squared Error (MSE)**

$$\text{MSE}(\hat{p}_2) = \text{Var}(\hat{p}_2) + [\text{Bias}(\hat{p}_2)]^2.$$

$$= \frac{5p(1 - p)}{72} + \left( \frac{1 - 2p}{12} \right)^2.$$

$$= \frac{5p(1 - p)}{72} + \frac{(1 - 2p)^2}{144}.$$

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                         |
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| 3. | <p>Let <math>X \sim N(\theta, 4)</math> where <math>\theta</math> is unknown. Consider the estimator:</p> $\hat{\theta} = 0.6X + 0.4d$ <p>where <math>d</math> is a constant.</p> <p>(a) Find the bias of <math>\hat{\theta}</math> as an estimator of <math>\theta</math></p> <p>(b) Calculate the variance of <math>\hat{\theta}</math> (c) Compute the MSE of <math>\hat{\theta}</math></p> <p>(d) For <math>\theta = 3</math> and <math>d = 2</math>, find the numerical MSE value</p> <p>We have <math>X \sim N(\theta, 4)</math> (so <math>E[X] = \theta</math>, <math>\text{Var}(X) = 4</math>) and the estimator</p> $\hat{\theta} = 0.6X + 0.4d,$ <p>with constant <math>d</math>.</p> <p><b>(a) Bias</b></p> $E[\hat{\theta}] = 0.6E[X] + 0.4d = 0.6\theta + 0.4d,$ <p>so</p> $\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta = 0.6\theta + 0.4d - \theta = 0.4(d - \theta).$ <p><b>(b) Variance</b></p> <p>Since <math>d</math> is constant and <math>\text{Var}(X) = 4</math>,</p> $\text{Var}(\hat{\theta}) = (0.6)^2 \text{Var}(X) = 0.36 \times 4 = 1.44.$ <p><b>(c) MSE</b></p> $\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2 = 1.44 + (0.4(d - \theta))^2 = 1.44 + 0.16(d - \theta)^2.$ <p><b>(d) Numerical value for <math>\theta = 3</math>, <math>d = 2</math></b></p> <p><math>\text{Bias} = 0.4(2 - 3) = -0.4</math> so <math>\text{Bias}^2 = 0.16</math>.</p> <p>Thus</p> $\text{MSE} = 1.44 + 0.16 = 1.60.$ | Mean Squared Error for Normal           |
| 4. | <p>A network monitor records the number of incoming packets per second over 5 seconds:</p> $7, 6, 8, 5, 4.$ <p>Assume number of packets per second follows <math>\text{Poisson}(\lambda)</math>.</p> <p>Find the MLE of <math>\lambda</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Maximum Likelihood Estimate for Poisson |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                |
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|    | $\hat{\lambda}_{MLE} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$ <p>The observed values are:</p> <p style="text-align: center;">7, 6, 8, 5, 4.</p> <p>Sum:</p> <p style="text-align: center;"><math>7 + 6 + 8 + 5 + 4 = 30.</math></p> <p>Sample size:</p> <p style="text-align: center;"><math>n = 5.</math></p> <p style="text-align: center;"> <math>\hat{\lambda} = \frac{30}{5} = 6.</math>  </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                |
| 5. | <p>A temperature sensor records CPU temperature (°C) once per minute over 6 minutes:<br/>14.8, 15.2, 14.5, 15.0, 14.7, 15.3.<br/>Assume the readings are i.i.d. Normal(<math>\mu</math>, <math>\sigma^2</math>).<br/>Find the MLEs of <math>\mu</math> and <math>\sigma^2</math>.<br/>Data: 14.8, 15.2, 14.5, 15.0, 14.7, 15.3 (<math>n = 6</math>)</p> <p>For i.i.d. <math>X_i \sim \mathcal{N}(\mu, \sigma^2)</math>:</p> $\hat{\mu}_{MLE} = \bar{X}, \quad \hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$ <p>Compute:</p> <ul style="list-style-type: none"> <li><math>\bar{X} = \frac{14.8 + 15.2 + 14.5 + 15.0 + 14.7 + 15.3}{6} = 14.9167.</math></li> <li><math>\hat{\sigma}^2 = \frac{1}{6} \sum (X_i - 14.9167)^2 \approx 0.07806.</math></li> </ul> <p>Answer:</p> <div style="border: 1px solid black; padding: 2px; display: inline-block;"> <math>\hat{\mu} = 14.9167, \quad \hat{\sigma}^2 = 0.07806</math> </div> | MLE for Normal |
| 6. | Let $X_1, \dots, X_n$ be a random sample from a $N(0, \sigma^2)$ population. Find the MLE of $\sigma$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Exercise 4.9   |

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|    | <p><b>Step 1: Likelihood function</b></p> <p>The pdf is</p> $f(x; \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma^2}\right).$ <p>For the sample:</p> $L(\sigma) = \prod_{i=1}^n f(x_i; \sigma) = \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^n \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2\right).$ <p><b>Step 2: Log-likelihood</b></p> $\ell(\sigma) = -n \ln \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2 - \frac{n}{2} \ln(2\pi).$ <p><b>Step 3: Differentiate</b></p> $\frac{d\ell}{d\sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^n x_i^2.$ <p>Set derivative = 0:</p> $-\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^n x_i^2 = 0 \quad \Rightarrow \quad \sum_{i=1}^n x_i^2 = n\sigma^2.$ <p><b>Step 4: Solve</b></p> $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n x_i^2, \quad \hat{\sigma} = \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2}.$ |                                       |
| 7. | <p>Find the value of <math>z_{\alpha/2}</math> to use in expression to construct a confidence interval with level:</p> <ol style="list-style-type: none"> <li>95%</li> <li>98%</li> <li>99%</li> <li>80%</li> </ol> <p>Ans:</p> <ol style="list-style-type: none"> <li>1.96</li> <li>2.33</li> <li>2.57 or 2.58</li> <li>d. 1.28</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Section 5.1</b><br><b>Q. No: 1</b> |
| 8. | <p>Find the levels of the confidence intervals that have the following values of <math>z_{\alpha/2}</math>:</p> <ol style="list-style-type: none"> <li><math>z_{\alpha/2} = 1.96</math></li> <li><math>z_{\alpha/2} = 2.17</math></li> <li><math>z_{\alpha/2} = 1.28</math></li> <li><math>z_{\alpha/2} = 3.28</math></li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Section 5.1</b><br><b>Q. No: 2</b> |

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|     | <p>Ans:</p> <ul style="list-style-type: none"> <li>a. 95%</li> <li>b. 97%</li> <li>c. 80%</li> <li>d. 99.9%</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                               |
| 9.  | <p>In a sample of 100 steel wires the average breaking strength is 50 kN, with a standard deviation of 2 kN</p> <ul style="list-style-type: none"> <li>. Find a 95% confidence interval for the mean breaking strength of this type of wire.</li> <li>. Find a 99% confidence interval for the mean breaking strength of this type of wire.</li> <li>. An engineer claims that the mean breaking strength is between 49.7 kN and 50.3 kN. With what level of confidence can this statement be made?</li> <li>. How many wires must be sampled so that a 95% confidence interval specifies the mean breaking strength to within <math>\pm 0.3</math> kN?</li> <li>. How many wires must be sampled so that a 99% confidence interval specifies the mean breaking strength to within <math>\pm 0.3</math> kN?</li> </ul> <p>Ans:</p> <p>(a) <math>X = 50, s = 2, n = 100, Z_{0.025} = 1.96</math>.<br/>The confidence interval is <math>50 \pm 1.96(2/\sqrt{100})</math>, or (49.608, 50.392).</p> <p>(b) <math>X = 50, s = 2, n = 100, Z_{0.005} = 2.58</math>.<br/>The confidence interval is <math>50 \pm 2.58(2/\sqrt{100})</math>, or (49.484, 50.516).</p> <p>(c) <math>X = 50, s = 2, n = 100</math>, so the upper confidence bound 50.3 satisfies <math>50.3 = 50 + z\alpha/2(2/\sqrt{100})</math>.<br/>Solving for <math>z\alpha/2</math> yields <math>z\alpha/2 = 1.50</math>.<br/>The area to the right of <math>z = 1.50</math> is <math>1 - 0.9332 = 0.0668</math>, so <math>\alpha/2 = 0.0668</math>.<br/>The level is <math>1 - \alpha = 1 - 2(0.0668) = 0.8664</math>, or 86.64%.</p> <p>(d) <math>z_{0.025} = 1.96</math>. <math>1.96(2/\sqrt{n}) = 0.3</math>, so <math>n = 171</math>.</p> <p>(e) <math>z_{0.005} = 2.58</math>. <math>2.58(2/\sqrt{n}) = 0.3</math>, so <math>n = 296</math>.</p> | <p><b>Section 5.1</b><br/><b>Q. No: 5</b></p> |
| 10. | <p>In a sample of 80 ten-penny nails, the average weight was 1.56 g and the standard deviation was 0.1 g.</p> <ul style="list-style-type: none"> <li>. Find a 95% confidence interval for the mean weight of this type of nail.</li> <li>b. Find a 98% confidence interval for the mean weight of this type of nail.</li> <li>c. What is the confidence level of the interval (1.54, 1.58)?</li> <li>d. How many nails must be sampled so that a 95% confidence interval specifies the mean to within <math>\pm 0.01</math> g?</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p><b>Section 5.1</b><br/><b>Q. No: 9</b></p> |

→ get 'Z' val  
→ find corresponding probability  
→  $\alpha$  = Probability  
 $1 - \alpha$  = confidence level

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                          |
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|     | <p>e. Approximately how many nails must be sampled so that a 98% confidence interval will specify the mean to within <math>\pm 0.01</math> g?</p> <p>Ans:</p> <p>(a) <math>X = 1.56</math>, <math>s = 0.1</math>, <math>n = 80</math>, <math>z_{0.025} = 1.96</math>.<br/>The confidence interval is <math>1.56 \pm 1.96(0.1/\sqrt{80})</math>, or (1.5381, 1.5819).</p> <p>(b) <math>X = 1.56</math>, <math>s = 0.1</math>, <math>n = 80</math>, <math>z_{0.01} = 2.33</math>.<br/>The confidence interval is <math>1.56 \pm 2.33(0.1/\sqrt{80})</math>, or (1.5339, 1.5861).</p> <p>(c) <math>X = 1.56</math>, <math>s = 0.1</math>, <math>n = 80</math>, so the upper confidence bound 1.58 satisfies <math>1.58 = 1.56 + z_{\alpha/2}(0.1/\sqrt{80})</math>.<br/>Solving for <math>z_{\alpha/2}</math> yields <math>z_{\alpha/2} = 1.79</math>.<br/>The area to the right of <math>z = 1.79</math> is <math>1 - 0.9633 = 0.0367</math>, so <math>\alpha/2 = 0.0367</math>.<br/>The level is <math>1 - \alpha = 1 - 2(0.0367) = 0.9266</math>, or 92.66%.</p> <p>(d) <math>z_{0.025} = 1.96</math>. <math>1.96(0.1/\sqrt{n}) = 0.01</math>, so <math>n = 385</math>.</p> <p>(e) <math>z_{0.01} = 2.33</math>. <math>2.33(0.1/\sqrt{n}) = 0.01</math>, so <math>n = 543</math>.</p> |                          |
| 11. | <p>In a sample of 60 electric motors, the average efficiency (in percent) was 85. Assume the population standard deviation is 2.</p> <p>a. Find a 98% lower confidence bound for the mean efficiency.</p> <p>b. The claim is made that the mean efficiency is greater than 84.6%. With what level of confidence can this statement be made?</p> <p>Ans.</p> <p>(a) Sample size <math>n=60</math>, Sample mean <math>\bar{x}=85</math>, Sample standard deviation <math>s=2</math>.<br/><math>\sigma_X \approx s/\sqrt{n} = 0.2582</math>.<br/>The 98% lower confidence bound is <math>\bar{x} - 2.05\sigma_X = 84.471</math>.</p> <p>(b)<br/>Confidence level = or 93.94%.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Section 5.1<br>Q. No: 11 |
| 12. | <p>In a simple random sample of 70 automobiles registered in a certain state, 28 of them were found to have emission levels that exceed a state standard.</p> <p>a) What proportion of the automobiles in the sample had emission levels that exceed the standard?</p> <p>b) Find a 95% confidence interval for the proportion of automobiles in the state whose emission levels exceed the standard.</p> <p>c) Find a 98% confidence interval for the proportion of automobiles whose emission levels exceed the standard.</p> <p>d) How many automobiles must be sampled to specify the proportion that exceed the standard to within <math>\pm 0.10</math> with 95% confidence?</p> <p>e) How many automobiles must be sampled to specify the proportion that exceed the standard to within <math>\pm 0.10</math> with 98% confidence?</p> <p>f) Someone claims that less than half of the automobiles in the state exceed the standard. With what level of confidence can this statement be made?</p> <p>Ans.</p> <p>(a) <math>28/70 = 0.4</math>, or 40%.</p> <p>(b) <math>X = 28</math>, <math>n = 70</math>, <math>\hat{p} = (28+2)/(70+4) = 0.405405</math>, <math>Z_{0.025} = 1.96</math>.</p>                                                                               | Section 5.3<br>Q. No: 1  |

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|     | <p>The confidence interval is <math>0.405405 \pm 1.96 \sqrt{(0.405405(1-0.405405)/(70+4))}</math>,<br/>or (0.294, 0.517).<br/>(c) <math>X = 28, n = 70, \hat{p} = (28+2)/(70+4) = 0.405405, Z_{0.01} = 2.33</math>.<br/>The confidence interval is <math>0.405405 \pm 2.33 \sqrt{(0.405405(1-0.405405)/(70+4))}</math>,<br/>or (0.272, 0.538).<br/>(d) Let <math>n</math> be the required sample size.<br/>Then <math>n</math> satisfies the equation <math>0.1 = 1.96 \sqrt{(\hat{p}(1-\hat{p})/(n+4))}</math>.<br/>Replacing with 0.405405 and solving for <math>n</math> yields <math>n = 89</math>.<br/>(e) Let <math>n</math> be the required sample size.<br/>Then <math>n</math> satisfies the equation <math>0.10 = 2.33 \sqrt{(\hat{p}(1-\hat{p})/(n+4))}</math>.<br/>Replacing with 0.405405 and solving for <math>n</math> yields <math>n = 127</math>.<br/>(f) The upper confidence bound 0.50 satisfies the equation<br/><math>0.50 = 0.405405 + z_{\alpha} \sqrt{(0.405405(1-0.405405)/(70+4))}</math><br/>Solving for <math>z_{\alpha}</math> yields <math>z_{\alpha} = 1.66</math>. The area to the left of <math>z = 1.66</math> is <math>1 - \alpha = 0.9515</math>.<br/>The level is 0.9515, or 95.15%.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                               |
| 13. | <p>High levels of blood sugar are associated with an increased risk of type 2 diabetes. A level higher than normal is referred to as “impaired fasting glucose.” The article “Association of Low-Moderate Arsenic Exposure and Arsenic Metabolism with Incident Diabetes and Insulin Resistance in the Strong Heart Family Study” (M. Grau-Perez, C. Kuo, et al., Environmental Health Perspectives, 2017, online) reports a study in which 47 of 155 people with impaired fasting glucose had type 2 diabetes. Consider this to be a simple random sample.</p> <p>a) Find a 95% confidence interval for the proportion of people with impaired fasting glucose who have type 2 diabetes.<br/>b) Find a 99% confidence interval for the proportion of people with impaired fasting glucose who have type 2 diabetes.<br/>c) A doctor claims that less than 35% of people with impaired fasting glucose have type 2 diabetes. With what level of confidence can this claim be made?</p> <p>Ans.<br/>(a) <math>X = 47, n = 155, \hat{p} = (47+2)/(155+4) = 0.3082, Z_{0.025} = 1.96</math>.<br/>The confidence interval is <math>0.3082 \pm 1.96 \sqrt{(0.3082(1-0.3082)/(155+4))}</math>, or<br/>(0.2365, 0.3799).<br/>(b) <math>X = 47, n = 155, \hat{p} = (47+2)/(155+4) = 0.3082, Z_{0.005} = 2.58</math>.<br/>The confidence interval is <math>0.3082 \pm 2.58 \sqrt{(0.3082(1-0.3082)/(155+4))}</math>, or<br/>(0.2137, 0.4027).<br/>© The upper confidence bound 0.35 satisfies the equation <math>0.35 = 0.3082 + z_{\alpha} \sqrt{(0.3082(1-0.3082)/(155+4))}</math><br/>Solving for <math>z_{\alpha}</math> yields <math>z_{\alpha} = 1.1421</math>. The area to the left of <math>z = 1.1421</math> is <math>1 - \alpha = 0.8729</math>.<br/>The level is 0.8729, or 87.29%..</p> | <p><b>Section 5.3</b><br/><b>Q. No: 5</b></p> |
| 14. | <p>To study the effect of curing temperature on shear strength of a certain rubber compound, 80 specimens were cured at 150°C and 95 were cured at 130°C. The specimens cured at 150°C had an average shear strength of 620</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p><b>Section 5.4</b><br/><b>Q. No: 1</b></p> |



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|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
|     | <p>psi, with a standard deviation of 20 psi. Those cured at 130°C had an average shear strength of 750 psi, with a standard deviation of 30 psi. Find a 95% confidence interval for the difference between the mean shear strengths of specimens cured at the two temperatures.</p> <p>Ans:<br/> <math>\bar{x}=620, s_X=20, n_X=80, \bar{y}=750, s_Y=30, n_Y=95, z_{0.025}=1.96</math>.<br/> <b>The confidence interval is <math>750-620 \pm 1.96 \sqrt{(20^2/80+30^2/95)}</math>, or (122.54, 137.46).</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                          |
| 15. | <p>In experiments to determine the effectiveness of drugs such as anti-fungal ointments that are applied to the skin, the concentration of the drug in a patient's skin must be measured. The article "Determining Bio-equivalence of Topical Dermatological Drug Products by Tape-Stripping" (W. Navidi, A. Hutchinson, et al., Journal of Pharmacokinetics and Pharmacodynamics, 2008:213–220) describes an improved method of measuring that is designed to ensure that all the drug in the skin is recovered. Assume that in an experiment using the old method, the mean amount of drug recovered from a sample of 49 patients was 105 ng with a standard deviation of 20 ng, and that in an experiment using the new method the mean amount of drug recovered in a sample of 35 patients was 117 ng with a standard deviation of 15 ng. Find a 98% confidence interval for the difference in the mean amounts recovered between the two methods.</p> <p>Ans:<br/> <math>\bar{x}=105, s_X=20, n_X=49, \bar{y}=117, s_Y=15, n_Y=35, z_{0.01}=2.33</math>.<br/> <b>The confidence interval is <math>117-105 \pm 2.33 \sqrt{(20^2/49+15^2/35)}</math>, or (2.934, 21.066).</b></p> | <b>Section 5.4</b><br><b>Q. No: 7</b>    |
| 16. | <p>Find the value of <math>t_{n-1, \alpha/2}</math> needed to construct a two sided confidence interval of the given level with the given sample size:</p> <ul style="list-style-type: none"> <li>a) Level 90%, sample size 12.</li> <li>b) Level 95%, sample size 7.</li> <li>c) Level 99%, sample size 2.</li> <li><b>d) Level 95%, sample size 29.</b></li> </ul> <p>Ans:<br/>           (a) 1.796<br/>           (b) 2.447<br/>           (c) 63.657<br/> <b>(d) 2.048</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Section 5.3</b><br><b>Q. No: 1</b>    |
| 17. | <p>The article "Wind-Uplift Capacity of Residential Wood Roof-Sheathing Panels Retrofitted with Insulating Foam Adhesive" (P. Datin, D. Prevatt, and W. Pang, Journal of Architectural Engineering, 2011:144–154) presents a study of the failure pressures of roof panels. Following are the failure pressures, in kPa, for five panels constructed with 6d smooth shank nails. These data are consistent with means and standard deviations presented in the article. 3.32 2.53 3.45 2.38</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Exercise 5.3</b><br><b>Question 5</b> |

3.01 Find a 95% confidence interval for the mean failure pressure for this type of roof panel.

Ans

Let the sample size be  $n = 5$ . The sample mean is:

$$\bar{x} = \frac{3.32 + 2.53 + 3.45 + 2.38 + 3.01}{5} = \frac{14.69}{5} = 2.938$$

The sample standard deviation is:

$$s = \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2} = \sqrt{\frac{0.89}{4}} = \sqrt{0.2225} \approx 0.4717$$

Degrees of freedom:  $df = n - 1 = 4$

For a 95% confidence level, the critical value is  $t_{4,0.025} = 2.776$

Now compute the margin of error:

$$ME = t \cdot \frac{s}{\sqrt{n}} = 2.776 \cdot \frac{0.4717}{\sqrt{5}} = 2.776 \cdot 0.2109 \approx 0.5855$$

The confidence interval is:

$$\bar{x} \pm ME = 2.938 \pm 0.5855 = (2.352, 3.524)$$

18.

In an accelerated failure test, components are operated under extreme conditions so that a substantial number will fail in a rather short time. In such a test involving two types of microchips, 600 chips manufactured by an existing process were tested, and 125 of them failed. Then 800 chips manufactured by a new process were tested, and 130 of them failed. Find a 90% confidence interval for the difference between the proportions of failures for chips manufactured by the two processes.

#### Step 1: Define the Given Values

Let:

- $x_1 = 125$  failures out of  $n_1 = 600$  chips from the existing process
- $x_2 = 130$  failures out of  $n_2 = 800$  chips from the new process

Calculate the sample proportions:

- $\hat{p}_1 = \frac{x_1}{n_1} = \frac{125}{600} \approx 0.2083$
- $\hat{p}_2 = \frac{x_2}{n_2} = \frac{130}{800} = 0.1625$

We are estimating the difference:

$$\hat{p}_1 - \hat{p}_2 = 0.2083 - 0.1625 = 0.0458$$

#### Step 2: Compute the Standard Error (SE)

The standard error for the difference in proportions is:

$$SE = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

$$SE = \sqrt{\frac{0.2083(1-0.2083)}{600} + \frac{0.1625(1-0.1625)}{800}} = \sqrt{\frac{0.1649}{600} + \frac{0.1361}{800}} = \sqrt{0.0002748 + 0.0001701} = \sqrt{0.0004449} \approx 0.0211$$

Difference of proportions

*Traditional method used. Use Agresti-Coull method only.*

**Step 3: Find the z-Critical Value**

For a 90% confidence level:

- $\alpha = 0.10$ , so  $\alpha/2 = 0.05$
- From the z-table,  $z_{0.05} = 1.645$

**Step 4: Compute the Margin of Error (ME)**

$$ME = z \cdot SE = 1.645 \cdot 0.0211 \approx 0.0347$$

**Step 5: Construct the Confidence Interval**

$$(\hat{p}_1 - \hat{p}_2) \pm ME = 0.0458 \pm 0.0347$$

The 90% confidence interval for the difference in failure proportions is:

$$(0.0111, 0.0805)$$

This suggests that the failure rate of the existing process is likely between 1.11% and 8.05% higher than that of the new process.

19. The article “Simulation of the Hot Carbonate Process for Removal of CO<sub>2</sub> and H<sub>2</sub>S from Medium Btu Gas” (K. Park and T. Edgar, Energy Progress, 1984:174–180) presents an equation used to estimate the equilibrium vapor pressure of CO<sub>2</sub> in a potassium carbonate solution. The actual equilibrium pressure (in kPa) was measured in nine different reactions and compared with the value estimated from the equation. The results are presented in the following table:

| Reaction | Estimated | Experimental | Difference |
|----------|-----------|--------------|------------|
| 1        | 45.10     | 42.95        | 2.15       |
| 2        | 85.77     | 79.98        | 5.79       |
| 3        | 151.84    | 146.17       | 5.67       |
| 4        | 244.30    | 228.22       | 16.08      |
| 5        | 257.67    | 240.63       | 17.04      |
| 6        | 44.32     | 41.99        | 2.33       |
| 7        | 84.41     | 82.05        | 2.36       |
| 8        | 150.47    | 149.62       | 0.85       |
| 9        | 253.81    | 245.45       | 8.36       |

Find a 95% confidence interval for the mean difference between the estimated and actual pressures.

Ans:

$\bar{D} = 6.736667$ ,  $s_D = 6.045556$ ,  $n = 9$ ,  $t_{9-1, 0.025} = 2.306$ .

The confidence interval is  $6.736667 \pm 2.306(6.045556/\sqrt{9})$ , or (2.090, 11.384).

**Section 5.7**  
**Q. No:**

20.

The article “Effect of Refrigeration on the Potassium Bitartrate Stability and Composition of Italian Wines” (A. Versari, D. Barbanti, et al., Italian Journal of Food Science, 2002:45–52) reports a study in which eight types of white wine had their tartaric acid concentration (in g/L) measured both before and after a cold stabilization process. The results are presented in the following table:

| Wine Type | Before | After | Difference |
|-----------|--------|-------|------------|
| 1         | 2.86   | 2.59  | 0.27       |
| 2         | 2.85   | 2.47  | 0.38       |
| 3         | 1.84   | 1.58  | 0.26       |
| 4         | 1.60   | 1.56  | 0.04       |
| 5         | 0.80   | 0.78  | 0.02       |
| 6         | 0.89   | 0.66  | 0.23       |
| 7         | 2.03   | 1.87  | 0.16       |
| 8         | 1.90   | 1.71  | 0.19       |

Find a 95% confidence interval for the mean difference between the tartaric acid concentrations before and after the cold stabilization process.

Ans:

$\bar{D}=0.19375$ ,  $s_{\bar{D}}=0.120230$ ,  $n=8$ ,  $t_{8-1,.025}=2.365$ .

The confidence interval is  $0.19375 \pm 2.365(0.120230/\sqrt{8})$ , or (0.0932, 0.2943).

Section 5.7

Q. No: 2